

face of manufacturing. Until now, the creation of high-quality physical products required expensive machinery and investments in tooling – and with developments such as low cost 3D modelling tools, investment websites and industry collaborations over designs, 3D modelling has the right canvas to make Do It Yourself (DIY) manufacturing a revolution in itself. While it is uncertain as to how 3D printing will exactly impact the traditional manufacturing sector, emerging trends indicate that something of a paradigm shift has already begun. One thing is for certain – as more and more organisations tend to become manufacturers, the line between consumer and manufacturer will blur. Few possible results of the emergence of 3D printing could be:

- ▶ **Products with superior capabilities:** Products incorporating 3D printed components will start exhibiting superior features such as being more durable, lighter, smaller, less mechanically complex and easier to maintain holding distinctive competitive advantage over traditional mass manufacturing units.
- ▶ **Products will have to be marketed quicker:** 3D printing would lead to faster design and faster prototyping cycles. Apart from this, the elimination of tooling and setup times of factories would all in unison mean that being quick to market would be less of a competitive advantage, and more of a necessity to survive in the quick paced industry.
- ▶ **Outsourcing might become moot:** The current advantages in terms of price when it comes to outsourcing or off shoring certain stages in a supply chain to low-cost regions might slowly become negligible thanks to the just-in-time manufacturing closer to the point of sale or assembly.
- ▶ **Designs will be open and community oriented:** Communities of end users will be responsible for various product designs that will be available to anyone with the skills and the tools required to manufacture them. These open designs can be modified and will be superior to proprietary products that are mass manufactured.
- ▶ **Customisation will become the norm:** As companies edge more and more towards using 3D printers and other technologies to offer customisation at minimal costs, customers will



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